



İZMİR EKONOMİ ÜNİVERSİTESİ
IZMIR UNIVERSITY OF ECONOMICS

STAJ RAPORU
INTERNSHIP REPORT

ÖĞRENCİNİN/ STUDENT'S

ADI SOYADI /

NAME-SURNAME : Kaoutar AMMARA

NUMARASI /

STUDENT ID NO : 20210615207

BÖLÜMÜ/

DEPARTMENT : AEROSPACE ENGINEERING

STAJ YAPTIĞI KURUMUN/ THE INSTITUTION'S

ADI/

NAME : Space Systems Design and Test Laboratory

ADRESİ/

**ADDRESS : Uzay Mühendisliği Bölümü, Uçak ve Uzay Bilimleri Fakültesi, İTÜ
34468, Maslak, İstanbul/ Türkiye**

STAJ YAPILAN BİRİM/

TÜRÜ/ : Research and Development (R&D)

DEPARTMENT/

KIND OF THE INTERNSHIP : Compulsory Internship

STAJ TARİHLERİ/ DATES OF INTERNSHIP

BAŞLAMA TARİHİ /

BEGINNING DATE : 11.08.2025

BİTİŞ TARİHİ /

COMPLETION DATE : 05.09.2025

Contents

SUMMARY	3
MAIN TEXT OF THE REPORT	3
QUESTION 1	3
QUESTION 2	4
QUESTION 3	4
QUESTION 4	5
QUESTION 5	5
QUESTION 6	6
QUESTION 7	6
INTERNSHIP REPORT	6
FIRST WEEK OF THE INTERNSHIP	6
SECOND WEEK OF THE INTERNSHIP	10
THIRD WEEK OF THE INTERNSHIP	15
FOURTH WEEK OF THE INTERNSHIP	20
References:	24

SUMMARY

During this internship with İTÜ Space Systems Design & Test Laboratory (USTTL/SSDTL), our work was relocated to office space at the TRISONIK Laboratory because the Aerospace Faculty building is undergoing seismic retrofitting; consequently, no clean-room or environmental-test access was possible this month. Orientation (per internal email) directed us to TRISONIK to bring laptops; we were informed that satellite development activities continue in a separate location and that environmental testing systems are currently inaccessible. Within these constraints, I focused on mission/facility research, STK Pro scenario building (visibility / access for Istanbul, preliminary lifetime framing), and strengthening my theoretical base in wireless communication and electromagnetics as applied to small-satellite missions. I also self-started on CATIA to understand industry-standard CAD workflows (and realized why CATIA/NX dominate complex aerospace assemblies compared to the more limited workflows I felt in SolidWorks). On the program side, I reviewed USTTL's heritage (e.g., ITUpSAT-1) and the latest missions: PAUSAT-1 (16U, launched 14 Jan 2025) and the RAFS 6U CubeSat (Rubidium Atomic Frequency Standard) targeting an early-2026 launch. These activities clarified subsystem interfaces, ground segment realities (via UHUZAM/CSCRS capabilities), and confirmed how essential STK is for my career.

QUESTION 1

Give the following information about the institution you performed your internship.

- a. Name and address of the institution

Name: İTÜ Space Systems Design & Test Laboratory (USTTL)

Address: İTÜ Ayazağa Kampüsü, Uçak ve Uzay Bilimleri Fakültesi, 34469, Maslak, İstanbul / Türkiye.

- b. Its field of activity

USTTL designs, integrates and tests pico/nano/micro-satellites and subsystems (ADCS, C&DH, COMMS, EPS, structures/thermal, optics) and conducts education/internship programs tied to flight projects and ground operations.

- c. Sort history of the institution that introduces the development of the institution

Founded within İTÜ's Faculty of Aeronautics & Astronautics to advance small-satellite capabilities, USTTL integrated and thermally qualified ITUpSAT-1, launched on PSLV-C14 (23 September 2009), establishing Türkiye's early CubeSat heritage. The lab has since contributed to multiple missions and, most recently, to PAUSAT-1 (16U) in collaboration with Pakistan Air University. In parallel, USTTL is developing the RAFS 6U CubeSat to flight-validate a Rubidium Atomic Frequency Standard for future national GNSS missions (project started mid-2023; PDR in 2024; launch targeted early-2026).

- d. Upper institution it is affiliated to (if any) and existing facilities

USTTL operates under İTÜ / Faculty of Aeronautics & Astronautics (UUBF). Facilities (temporarily inaccessible during retrofitting) include:

Class 1000 clean room (~25 m²) and Class 100,000 clean area (~8 m²).

Thermal Vacuum Chamber (TVAC).

Vibration Test System.

Electromagnetic Diagnostic & Measurement Laboratory.

Rapid Prototyping Lab.

LPKF PCB plotter.

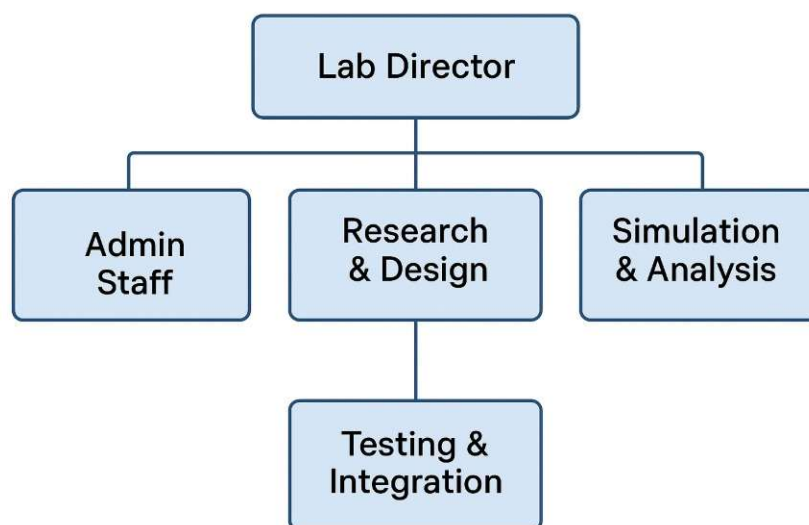
Additionally, İTÜ UHUZAM (CSCRS) provides ground segment infrastructure: 13 m remote-sensing antenna and 2.4 m / 4.6 m VSAT antennas, with X-band reception coverage up to ~3,000 km diameter (from Sweden to Sudan; England to Kazakhstan).

- e. Number of employees in the institution and their classification according to their qualities (worker, technician, engineer, administrative personnel, etc.)

As an academic laboratory, staffing fluctuates by term. The active team typically includes a Principal Investigator (Professor), Project/Sub-system leads, Research Assistants/Graduate Researchers, and Undergraduate interns; Faculty administration provides shared services (procurement, compliance, scheduling). (Exact counts vary each semester.)

QUESTION 2

By drawing the organizational chart of the institution; please state which units fulfill production/service and general business administration functions, relations, authority and responsibilities between the departments.



QUESTION 3

Did you have a chance to use your foreign language during your internship? What was this foreign language? Please answer in Yes/No format and specify which languages you used.

Yes. I used English for technical resources (STK documentation, academic papers) and Turkish/English for local communications and procedural notes.

QUESTION 4

State your ideas and suggestions that could increase efficiency in the processes of production and service at the institution you do the internship.

In my opinion, efficiency could be improved by encouraging engineers to spend more time regularly explaining their tasks to interns. This would create more opportunities for interaction, helping interns better connect their academic studies with real applications while feeling more engaged as part of the team. In addition, a structured onboarding process - such as a one-page “welcome” note, a buddy system, and introductory tutorials- would help newcomers adapt quickly with fewer disruptions. Finally, expanding access to remote tools, automating repetitive ground station tasks, and maintaining a digital archive of past projects could save time and improve continuity across both research and service processes.

QUESTION 5

Please list the duties you performed throughout your internship and explain what you did about these duties. Please attach the visual documents if any to the Internship Report.

- During my internship, I began with an orientation at the TRISONIK Lab, where I was informed about the seismic retrofitting of our main building and the temporary suspension of environmental testing. Since we could not access the clean rooms or test facilities, I focused on researching USTTL’s infrastructure, including the clean rooms, thermal vacuum chamber, vibration system, electromagnetic lab, rapid prototyping lab, and the UHUZAM ground segment with its 13-meter antenna and VSAT systems. One of the main tasks I was given was to study wildfire detection sensors. Under my professor’s guidance, I researched OroraTech’s SAFIRE-Gen2 sensor, learning how it achieves 200 m/pixel GSD and wide swath coverage with an uncooled microbolometer and tri-band filters. I compared this with other cooled and uncooled systems, noting their trade-offs in power, resolution, and sensitivity, and we were asked to think about designing something similar for a CubeSat. Following this, I worked on STK Pro, where I built access and visibility scenarios for Istanbul using TLEs and SGP4, and later started modeling a wildfire detection payload by defining sensor footprints and coverage over Türkiye. Along the way, I also gained awareness of USTTL’s heritage projects, such as ITUpSAT-1, and their current missions like PAUSAT-1 and RAFS 6U. To strengthen my design skills, I taught myself the basics of CATIA (parts, assemblies, constraints) and reflected on why it is widely used in aerospace compared to SolidWorks, which I found less effective for complex assemblies. Finally, I revisited my knowledge in communications and electromagnetics, especially Friis link budgets, antenna fundamentals, and propagation, and connected these directly to how UHUZAM receives and processes satellite data.

QUESTION 6

What kind of applications did you see about the theoretical knowledge you gained? Specify openly. Give an example.

- I was able to apply different parts of my theoretical knowledge to practical work. My strongest area was in wireless communications and electromagnetics, where I used concepts like the Friis link budget and antenna gains to understand how downlinks from CubeSats can be supported by ground stations such as UHUZAM. This helped me connect classroom theory directly to real-world satellite operations. On the other hand, I realized that my background in orbital mechanics was limited, which made it challenging at first to work with orbital lifetime and access analysis in STK. Still, I was grateful to the professor for giving us a book called “Orbital Mechanics For Engineering Students” by Howard D. Curtis, it was fairly handy for us I was able to learn by doing: I imported TLEs, generated access windows over Istanbul, and experimented with how changing the drag and area-to-mass ratio affects orbital lifetime. A major highlight was when we tried to model a wildfire detection payload in STK, where I attached a sensor with parameters inspired by OroraTech’s SAFIRE-Gen2 and simulated its footprint and swath coverage over Türkiye. This linked my optics knowledge (focal length, pixel pitch, GSD formula) with orbital analysis and gave me a clearer picture of how satellites detect and monitor natural disasters

QUESTION 7

Please tell what this internship gave to you in one paragraph.

- This internship gave me much more than I expected, even though we did not have access to the main laboratory because of the seismic retrofitting. It showed me how different subsystems and facilities come together in satellite projects, and it helped me understand how important the ground segment is for mission success. I am grateful that I could learn STK, because I now see it as a crucial tool in aerospace engineering, and I even wish it had been introduced earlier in my studies. I also became more aware of my own strengths and weaknesses: I was happy with how solid my wireless communication knowledge felt, but I also realized that I need to improve in orbital mechanics. On the design side, teaching myself CATIA was both challenging and rewarding. What stood out most for me was working on the wildfire detection project, where I got to study OroraTech’s approach and imagine how we could design something similar. Even though I sometimes felt frustrated by the restrictions, I left this internship feeling more motivated, better prepared for real engineering applications, and genuinely excited about contributing to future satellite missions like RAFS 6U

INTERNSHIP REPORT

FIRST WEEK OF THE INTERNSHIP

In the first week of my internship, I did not have the chance to enter the clean rooms or laboratories due to the ongoing seismic retrofitting of the Aerospace Faculty building. Instead, I spent the week conducting comprehensive research on the lab’s facilities, its previous CubeSat missions, and key satellite subsystems. The week concluded with a written report summarizing my findings, which became my first deliverable for the internship.

1. Research on USTTL Facilities

My initial focus was on understanding the infrastructure available at USTTL, which plays a central role in CubeSat development. Although access was restricted, I studied each facility's purpose and specifications:

Clean Rooms:

Class 1,000 Clean Room (~25 m²) — used for spacecraft integration and thermal vacuum testing.

Class 100,000 Clean Room (~8 m²) — mainly for educational and demonstration purposes.

Testing Facilities:

Thermal Vacuum Chamber (TVAC): Capable of simulating deep-space vacuum (~10⁻⁶ torr) and temperature extremes between -20 °C and +60 °C.

Vibration Test System: Used to replicate launch stresses with random vibration profiles up to 2,000 Hz (Grms ≈ 8–12).

Electromagnetic Diagnostic Laboratory: For EMC/EMI testing, ensuring compliance with MIL-STD-461 and ECSS standards.

Prototyping and Electronics:

Rapid Prototyping Laboratory for structural parts.

LPKF PCB Plotter for custom circuit board fabrication.

In addition, I researched UHUZAM (İTÜ CSCRS), the university's ground station center.

UHUZAM operates a 13-meter remote sensing antenna capable of receiving X-band data over a 3,000 km coverage circle (spanning from Sweden to Sudan and England to Kazakhstan). This capability is supported by 2.4 m and 4.6 m VSAT antennas, ensuring continuous telemetry, tracking, and command (TT&C) coverage.

2. USTTL CubeSat Missions

The second part of my research focused on the CubeSat projects led or co-developed by USTTL, tracing the evolution of Turkish CubeSat technology:

İTÜpSAT-1 (2009): Türkiye's first CubeSat, a 1U nanosatellite launched on PSLV-C14. It tested passive attitude control and carried a CMOS imager. Though its design life was 6 months, it operated for over a year.

TurkSat-3USat (2013): A 3U CubeSat (~4 kg) with an amateur-band communication payload and passive stabilization. It included a self-deorbit device.

BeEagleSat & Havelsat (2017): 2U CubeSats developed as part of the international QB50 program. BeEagleSat carried an X-ray detector (XRD), while Havelsat focused on ionospheric research.

UBAKUSAT (2018): A 3U CubeSat deployed from the ISS in collaboration with JAXA, featuring amateur radio communication and radiation experiments.

ASELSAT (2021): A 3U CubeSat demonstrating an X-band transmitter and an optical camera.

PAUSAT-1 (2025): A 16U CubeSat launched on Falcon 9 (14 January 2025), carrying a multispectral imager with ~1.5 m GSD for Earth observation.

RAFS CubeSat (Planned 2026): A 6U satellite carrying a Rubidium Atomic Frequency Standard, intended to validate spaceborne atomic clock performance for future GNSS applications.

Project Name	Years	Institution	Mission	Size	Status	Orbital Deployment Date and Location	Payload
ITUpSAT1	2005 - 2010	ITU, TUBITAK	Training, system development, satellite data acquisition, passive orientation	1U	The status of the ITUpSAT1 is still active in the year of 2025.	September 23, 2009 from Satish Dhawan Space Centre.	Position sensing and stability via gyros, accelerometers, magnetometers, and AlNiCo magnet, Low-res (640×480) CMOS camera for imagery
TURKSAT 3U	2010 - 2013	TURKSAT	Communication, local system and subsystem development, education, inter-institutional collaboration	3U	Nanosatellite is still active in the year of 2025.	26th of April, 2013 from Jiuquan Satellite Launch Center, China	VHF/UHF linear transponder for voice communication.
Havelsat	2012 - 2017	HAVELSAN, QB50	Ionospheric science, international and national cooperation, system and subsystem development	2U	Havelsat completed its orbital life in March 2019.	May 16, 2017 Released from the ISS into orbit (USA, Nanoracks)	Multi-needle Langmuir probe for ionospheric electron density measurement.
BeEagle Sat	2012 - 2017	HHO/HUTEN and SU jointly, QB50 and Havelsan support	Ionospheric science, domestic system, subsystem, and sensor development, international and national cooperation	2U	Completed its orbital lifecycle at the end of April 2019.	May 26, 2017 Released from the ISS into orbit (USA, Nanoracks)	Multi-needle Langmuir probe (mNLP) for electron density measurement. CdZnTe-based hard X-ray detector (XRD) for 20–150 keV background characterization in LEO.
UBAKUSAT	2014 - 2018	HUTGM/U DHB, Japan /JAXA, TAMSAT,	Communication, passive orientation, analysis,	3U	Completed its mission in December	It was released into orbit from the ISS on	Amateur radio transponder for global voice communications.

		Gumush, and Ertek Uzay (project partners)	international and national cooperation, and domestic subsystem use		2020 when it reentered Earth's atmosphere .	May 11, 2018 (by Japan, JAXA).	
ASELSAT	2016 - 2019	ASELSAN	X-band communication Technology demonstration, domestic subsystem use	3U	ASELSAT completed its mission by reentering Earth's atmosphere in early 2025.	Launched on January 25, 2021 Falcon 9 Transporter	Experimental mini X-band transmitter with patch antenna (ASELSAN)
PIRISAT	2018 – 2020	STM	Automatic Identification System (AIS), Langmuir probe, radiation measurement	6U	Flight model preparations	Its launch date is still uncertain	Space-based AIS for modular hosting of third-party payloads.
LAGARI	2017 – 2020	STM	High-resolution earth observation	Microsatellite (50*50*70cm)	Flight model preparations	Its launch date is still uncertain	HRVI-2HD for 1.93 m Earth observation (500 km orbit), supporting reconnaissance, mapping, agriculture, forestry, and disaster monitoring.
Connect a t1.1	2021 – 2022	Plan S	Technology demonstration	3U	It is still working at the Earth's orbit in 2025.	25 May, 2022	Software-defined radio for bidirectional M2M/IoT communication trials.
Sharjah Sat 1	2019 - 2022	UoS, SU	Earth observation and X-ray detector	3U	It is still working at the Earth's orbit in 2025.	3 January 2023	Improved X-ray detector (iXRD) and dual cameras for scientific research and Earth observation.
PAUSAT 1	2022 - 2025	Pakistan Air University	1.5 m GSD 8 Band MS ground observation	16U	It is still working at the Earth's orbit in 2025.	Launched on January 14, 2025 Falcon 9 Transporter	High-res multispectral imager (1.5 m) and hyperspectral imager for detailed Earth observation.
RAFS	2024 - 2026	TUA and TUBITAK UME	RAFS Atomic Clock Testing in Low-Earth Orbit	6U	Engineering model being prepared	Planned for launch in the first half of 2026, the Falcon 9 Transporter	Atomic clock (RAFS) for testing in LEO for future national GNSS satellites (TUA project).

Table 1: form factor, launch date, and mission purpose for each CubeSat.

3. Learning New Subsystems and Technical Terms

While researching, I encountered many subsystem names and acronyms that I have not studied before. I took this chance to expand my technical vocabulary and ensure I understood their functions:

ADCS (Attitude Determination and Control System): Ensures correct satellite pointing using sensors (sun sensors, star trackers, gyros, GPS) and actuators (reaction wheels, magnetorquers).

EPS (Electrical Power System): Solar panels, batteries, and power distribution circuits.

OBC (On-Board Computer): Central data handling and fault detection system.

TTC (Telemetry, Tracking, and Command): Provides uplink and downlink for communication with the ground.

Payload Integration: The process of mechanically and electrically integrating mission-specific sensors.

Fig : A block diagram of a CubeSat bus showing ADCS, EPS, OBC, TTC, and Payload subsystems.

4. Weekly Report and Reflections

At the end of the week, I compiled my findings into a structured research report, which summarized the lab's facilities, its missions, and the technical subsystems of CubeSats. Writing the report helped me consolidate all the information I had gathered, and also forced me to revisit many new terms until I was comfortable with their meanings.

Looking back, this first week was essential in giving me a broad overview of USTTL's capabilities and history. Even though I was disappointed not to physically enter the clean rooms or testing areas, I still gained valuable knowledge about the technologies and systems I will be working with in the coming weeks.

SECOND WEEK OF THE INTERNSHIP

The second week of my internship was particularly exciting because it was the first time I engaged directly with the study of thermal sensors for wildfire detection. Our professor tasked us with researching existing satellite payloads, with a strong emphasis on OroraTech's SAFIRE-GEN2 sensor suite. This marked my first structured experience in conducting deep online technical research, which I found both challenging and rewarding, as it taught me how to track down precise technical specifications from scientific papers, conference presentations, and manufacturer datasheets.

1. Learning Through Research

At the beginning of the week, I was introduced to the technical background of wildfire detection sensors. I quickly realized that this work would not only improve my knowledge of satellite payloads but also sharpen my academic research skills. I learned how to identify reliable sources, navigate scientific databases, and extract key specifications from highly detailed documents. Many new technical words and concepts appeared during this process, and

I dedicated time to defining and explaining each of them in my notes. This practice will be especially useful in my future research projects and eventually my thesis work.



**ORORA
TECHNOLOGIES**

Thermal Infrared Imager Development for CubeSat Constellation



TUM

Abstract

Wildfires are becoming a major challenge to our civilization today. Early warning and detection of wildfires not only helps in managing them but also mitigates the loss of lives and property. To meet this challenge, OroraTech is developing a constellation of CubeSats for wildfire detection to alert about wildfire hazards within minutes instead of hours.

Onboard these CubeSats, a thermal infrared imager will scan the surface in multiple spectral bands to identify fires. Radiometric simulations with influences of sun glint, daylight and atmospheric absorption were done. A prototype of the imager based on an uncooled micro-bolometer focal plane array was developed including a shutter system, thermal stabilization, mounting structure and a data processing unit. The imager was tested for absolute temperature accuracy and noise behavior. With image processing the performance of the imager was further improved. The prototype is scheduled to fly on a stratospheric balloon late 2020, and the on-orbit demonstration is planned for early 2021.



Figure 1: Image of an active wildfire.

Radiometric Simulation

To prove the feasibility of the infrared imager, the conditions were modelled including:

- Radiation from soil temperature
- Atmospheric absorption
- Diffuse reflection of the sun
- Specular reflections of the sun
- Small sized fires

The results are plotted in Figure 5.



Figure 4: Illustration of the effects included in the radiometric simulation.

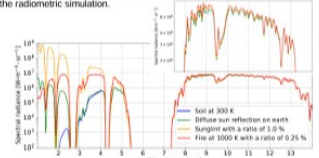


Figure 5: The simulated spectral radiance expected in space with a comparison of the effects included in the simulation. In mid wave infrared the sun glint dominates, in long wave infrared the emission by the fire is higher than sun glint.

Experiments

To test the performance of the setup built, a set of experiments were conducted. In the following list, the experiments are presented along with their results.

- Noise behavior:**
 - Evaluated by measuring the NETD (noise equivalent temperature difference)
 - NETD = 100 mK
- Absolute temperature accuracy**
 - Measured known surface at 4 m distance
 - Sample image shown in Figure 8
 - Maximum deviation of 2.5 K
 - Deviation due to size of source effect
- Impact of detector temperature**
 - Noise increases with higher temperature
 - Best performance measured at 20°C
- Sub-pixel hotspot detection**
 - Black body source behind an aperture
 - Object at the hyperfocal plane, 10 m
 - Aperture area at 1% of the pixel with a temperature difference of 25 K detectable
 - Figure 9 shows a detected sub-pixel hotspot



Figure 8: Infrared image obtained during the temperature accuracy test.

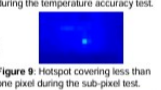


Figure 9: Hotspot covering less than one pixel during the sub-pixel test.

Motivation and Theory



Figure 2: Spatial & temporal resolution of existing satellites with thermal imaging bands and OroraTech's constellation. [1]

Objects with a temperature higher than absolute zero emit radiation as per Planck's law. Thus, the emission spectra of an ideal black body can be calculated from its temperature as shown in Figure 3. Note the shift of emissions peak into mid-wave infrared for hotter objects from room temperatures.

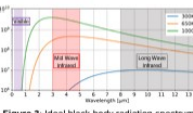


Figure 3: Ideal black body radiation spectrum.

Infrared Imager Setup

The infrared imager built consists of three main components. The shutter used for off-set correction, the optics and the detector which is an uncooled micro-bolometer. The whole setup is depicted in Figure 6.

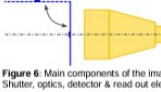


Figure 6: Main components of the imager (l. to r.): Shutter, optics, detector & read out electronics.

Even though it is uncooled, the temperature of the detector has a significant impact on the measurements. Thus a thermal control system was developed to maintain the detector at a stable operating temperature achieving 8 mK standard deviation. See stabilization at 35°C in Figure 7.

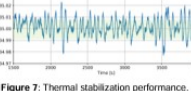


Figure 7: Thermal stabilization performance.

Conclusion and Outlook

Conclusion of the work:

- Radiometric simulation conducted including multiple expected effects
- Infrared imager developed and built. Figure 10 shows a photo of the prototype
- Experiments conducted to characterize the performance of the infrared imager
- Requirements for a possible space mission fulfilled

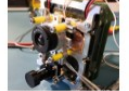


Figure 10: Photo of infrared imager.

The short-term goal is to fly the system on a stratospheric balloon later this year. With the experience gained from this mission, further development of the system are planned and will finally fly on satellites forming a constellation to achieve the temporal resolution target. An example Walker-Star constellation of 96 satellites is shown in Figure 11.



Figure 11: Possible constellation. satellites is shown in Figure 11.

References

[1] S. B. Shah, T. Grubler, L. Krempel, S. Ernst, F. Mauracher, and S. Contractor. Real Time Wildfire Detection from Space: A Trade-Off Between Sensor Quality, Physical Limitations & Payload Size. (Poster), ISPRS, XLII-2/W16.209-213, 2019.

Lucas Krempel^{1,2}, Prof. Reinhard Kienberger², Sanket B. Shah¹, Thomas Grubler¹

¹Orbital Oracle Technologies GmbH, Agnes-Pockels-Bogen 1, 80992 Munich, Germany

²Technische Universität München, Arcisstraße 21, 80333 Munich, Germany

* Contact: lucas.krempel@tum.de

SSC20-WP2-11

OroraTech technical presentation to illustrate how you collected data.

-> Key definitions I learned this week:

- NETD (Noise Equivalent Temperature Difference): The smallest temperature difference a sensor can detect; a lower NETD means higher sensitivity.
- Microbolometer: An uncooled infrared detector that measures radiation as a change in resistance due to heating.
- Stirling Cooler: A cryogenic device used in cooled detectors to reduce thermal noise.
- MWIR (Mid-Wave Infrared): Spectral range from 3–5 μm , useful for detecting high-temperature events like fires.
- LWIR (Long-Wave Infrared): Spectral range from 8–14 μm , more sensitive to background thermal radiation and Earth's surface.

2. OroraTech's SAFIRE-GEN2 Sensor

Through my research, I gathered detailed specifications of OroraTech's SAFIRE-GEN2 sensor, which is widely used in their wildfire monitoring CubeSats. Some of the key specifications I reported to my professor included:

Detector: Uncooled microbolometer.

Sensor: 1024×728 pixels, $17 \mu\text{m}$ pixel pitch.

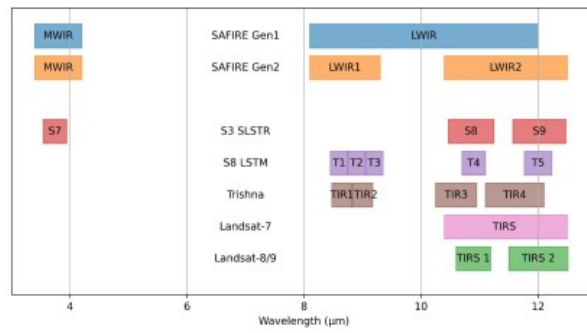
Optics: 50 mm focal length, $f/0.72$ aperture.

Spectral Ranges: Three bands — 3.8 μm (MWIR), 8.8 μm (LWIR), and 11.5 μm (LWIR).

Sensitivity (NETD): $< 50 \text{ mK}$.

Performance: Swath of $\sim 400 \text{ km}$ and Ground Sample Distance (GSD) of $\sim 200 \text{ m/pixel}$ at orbital altitudes.

SAFIRE Gen 2



Detector	Uncooled microbolometer
GSD	200 m
Swath	400 km
MWIR	3.4 – 4.2 μm
LWIR1	8.1 – 9.3 μm
LWIR2	10.4 – 12.5 μm

Figure : SAFIRE-GEN2 sensor parameters (detector type, optics, NETD, GSD, swath)

GSD formula used in our study:

$$GSD = (H \times p) / f$$

where H is the satellite altitude (m), p is the detector pixel pitch (m), and f is the focal length of the optics (m).

For SAFIRE-GEN2: H=550,000 m , p=17×10⁻⁶ m , f=0.05 m.

Calculation:

$$GSD = 550,000 \times 17 \times 10^{-6} / 0.05 \approx 187 \text{ m/pixel}$$

This result closely matches OroraTech's specification of ~200 m/pixel, which confirmed that our understanding of the formula was correct.

Learning about these numbers helped me understand how relatively compact CubeSats can still achieve large-scale Earth observation tasks such as wildfire detection.

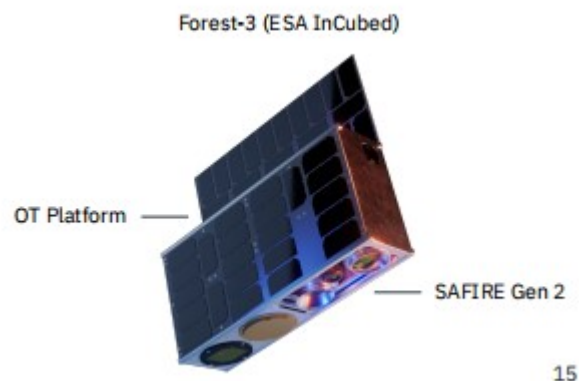


Figure : Forest-3 Satellite with SAFIRE Gen-2

3. Cooled vs. Uncooled Sensor Options

After collecting the baseline data from OroraTech's SAFIRE-Gen2, our next task was to compare two candidate payload designs: one based on a cooled infrared system and the other based on an uncooled system. This process involved extracting data from technical papers, conference presentations, and component datasheets, then organizing it into a structured format so we could evaluate trade-offs.

Cooled System (Leonardo Condor HD DWIR)

Detector: Leonardo Electronics Condor HD, a dual-waveband MWIR (3–5 μm) and LWIR (8–10 μm) detector.

Resolution & Pixel Pitch: 1280 $\times$ 1024 pixels (SXGA) with 12 μm pitch, which is significantly finer than uncooled options.

Sensitivity (NETD): 12 mK (MWIR), 26 mK (LWIR) — extremely sensitive, allowing early flame detection and better land surface temperature (LST) measurements.

Optics:

- Edmund Optics 25.4 mm diameter $\times$ 35 mm focal length.
- ZnSe Plano-Convex lens, with 3–12 μm broadband AR coatings.
- Filters: Iridian spectral mosaic filters at 3.8 μm , 8.8 μm , 10.1 μm .
- Achieved GSD $\approx$ 188 m/pixel at 550 km, swath $\approx$ 433 km (using 2 IR cameras with 10% overlap).

RGB Context Camera: SIMERA SENSE TriScape100, 4096 $\times$ 3072 pixels, 4.75 m GSD at 500 km.

Electronics: Jetson Xavier NX processor, FPGA front-end, mechanical shutter.

Power Consumption: 33–40 W (including cooler, FPGA/NUC, processor).

Uncooled System (BIRD 640 VOx Microbolometer)

Detector: SDC BIRD 640 ; VOx microbolometer, TEC-less (no cooling).

Resolution & Pixel Pitch: 640 $\times$ 480 pixels (VGA) with 17 μm pitch.

Sensitivity (NETD): < 35 mK (IR), < 22 mK (LWIR).

Optics:

- Edmund Optics 50 mm diameter $\times$ 50 mm focal length.
- Ge Plano-Convex lens, dual-band AR coatings.
- Filters: Iridian mosaic filters at 3.8 μm , 8.8 μm , 10.1 μm .
- Achieved GSD $\approx$ 187 m/pixel at 550 km, swath $\approx$ 214 km (using 2 IR cameras with 10% overlap).

RGB Context Camera: Same as cooled option (TriScape100).

Electronics: Jetson Xavier NX processor, FPGA front-end, internal shutter.

Power Consumption: 10–20 W total (detector power < 0.32 W).

-----How We Determined the Optimal Choices?-----

Our process was essentially a trade-off study:

Step 1 – Extract Specifications: We gathered specs for each candidate (detectors, optics, electronics, RGB).

Step 2 – Apply GSD Formula: $GSD = (H \times p)/f$

For cooled (12 μm pitch, 35 mm focal length):

$$IFOV = \frac{\text{pixel pitch}}{\text{focal length (rad)}} = \frac{12 \times 10^{-6}}{0.035} = 343 \text{ rad}$$

$$GSD(@550\text{km}) = H \times IFOV \text{ (m)} = 550000 \times 340 \times 10^{-6} = 188 \text{ m/pixel}$$

$$\text{Swath} = N \times GSD = 1280 \times 187 = 241 \text{ km}$$

For uncooled (17 μm pitch, 50 mm focal length):

$$IFOV = \frac{\text{pixel pitch}}{\text{focal length (rad)}} = \frac{17 \times 10^{-6}}{0.05} = 340 \mu\text{rad}$$

$$GSD(@550\text{km}) = H \times IFOV \text{ (m)} = 550000 \times 340 \times 10^{-6} = 187 \text{ m/pixel}$$

$$\text{Swath} = N \times GSD = 640 \times 187 = 119 \text{ km}$$

This confirmed that both systems can meet the ~ 200 m/pixel requirement.

Step 3 – Compare Swath & Coverage: The cooled system provided almost double the swath (433 km vs. 214 km).

Step 4 – Evaluate Power & Mass: The uncooled system was clearly more CubeSat-friendly (10–20 W vs. 33–40 W; <1 kg vs. several kg including cooler).

Step 5 – Report Trade-Offs: We presented both options to our professor, showing that the cooled system is ideal for high sensitivity, dual-band performance, while the uncooled system is more practical for CubeSat integration.

Parameter	Cooled System (Condor HD DWIR)	Uncooled System (BIRD 640 VOx)
Detector	MWIR+LWIR, SXGA, 12 μm pitch	3–14 μm , VGA, 17 μm pitch
Resolution	1280 $\times$ 1024 (SXGA)	640 $\times$ 480 (VGA)
NETD	12 mK (MWIR), 26 mK (LWIR)	< 35 mK (IR), < 22 mK (LWIR)
Optics	35 mm FL, ZnSe lens + filters	50 mm FL, Ge lens + filters
GSD @ 550 km	~ 188 m/pixel	~ 187 m/pixel
Swath Width	433 km (dual camera overlap)	214 km (dual camera overlap)
Power Consumption	33–40 W	10–20 W
Mass	< 750 g detector + cooler system	7 g detector + minimal electronics
Cooling Requirement	Stirling cryocooler	None
Pros	Dual-band, very high sensitivity, wide swath, high	Lightweight, simple, low power, CubeSat-friendly

Parameter	Cooled System (Condor HD DWIR)	Uncooled System (BIRD 640 V0x)
	res	
Cons	Heavy, complex, radiator area needed, shorter lifetime	Lower res, weaker MWIR, higher NETD

Figure : A side-by-side comparison chart of “Cooled vs. Uncooled” systems

4. Expanding My Technical Vocabulary

Throughout this week, I encountered and learned new technical terms such as NETD (Noise Equivalent Temperature Difference), microbolometer, Stirling cooling engine, MWIR (Mid-Wave Infrared), and LWIR (Long-Wave Infrared). Understanding these terms allowed me to better interpret sensor datasheets and technical papers, and it gave me confidence in reading advanced research materials that I had initially found overwhelming.

5. Weekly Report and Reflection

We ended the week by preparing and submitting a detailed report to our professor, summarizing our findings and presenting the two sensor options (cooled and uncooled). This was the first time I had the responsibility of compiling real-world technical research into a professional-style report, and it felt like a major step forward for me. I especially enjoyed the process of comparing specifications across systems and understanding how companies like OroraTech optimize their designs. Most importantly, I realized that I truly enjoy research work: the process of searching, reading, synthesizing, and reporting is something I now see as a strength that I will carry into my academic career and future theses.



Figure : 2nd week’s report presentation on thermal sensor choices

THIRD WEEK OF THE INTERNSHIP

Visit to UHUZAM and Satellite Pass Observation

During the third week, we visited UHUZAM/CSCRS (Center for Satellite Communications and Remote Sensing), where I witnessed my first orbital pass of PAUSAT-1. This was one of the highlights of my internship, as I could directly see how ground stations track satellites and handle live signals.

In orbital mechanics, a pass is defined by its AOS (Acquisition of Signal) and LOS (Loss of Signal):

AOS is the moment when the satellite first rises above the ground station’s horizon and its signal becomes detectable. LOS is the moment when the satellite sets and communication ends.

The pass duration can be estimated using the formula:

$$t_{pass} = \frac{2}{\omega} \cdot \cos^{-1} \left(\frac{R_E}{R_E + h} \cdot \cos(E_{min}) \right)$$

where:

ω = Earth's angular velocity (~15°/min for LEO)

R_E =6371 km (Earth radius)

h =550 km (satellite altitude)

For PAUSAT-1 at 550 km, with a minimum elevation of 20°, I calculated the pass duration to be about 9 minutes. The pass we witnessed was ~17° elevation, slightly below the decoding threshold, which explained why the signal on the spectrum analyzer was visible but not strong enough to fully decode telemetry.



Figure . PAUSAT-1 downlink detected during orbital pass (visible as a spectral peak around 2.2 GHz)

Ground Station Infrastructure

We were shown the large dish antenna outside UHUZAM, which provides X-band reception with a coverage footprint of ~3,000 km. Inside the building, we saw the stacked rack-mounted computers and receivers that form the backbone of the ground segment. These racks included demodulators, spectrum analyzers, servers, and signal processors. Their roles can be summarized as:

RF Front-End: Filters and amplifies the incoming signal from the antenna.

Downconverters: Translate high-frequency X-band/S-band signals to intermediate frequencies for easier processing.

Spectrum and Vector Network Analyzer (SVNA): Monitors the spectral profile, displaying carrier frequency peaks and Doppler shift in real-time.

Servers/Computers: Handle demodulation, decoding, telemetry extraction, and data storage.

Spectrum and Vector Network Analyzer (SVNA)

The SVNA is a hybrid instrument that combines spectrum analysis with vector network analysis. For satellites, the spectrum analyzer mode is most important, as it displays frequency vs. signal power (dBm), showing the satellite's carrier as a peak above the noise floor.

Signal Power Equation (Friis Transmission): $P_r = P_t + G_t + G_r - L_p - L_{sys}$

where:

P_r = received power (dBm)

P_t = transmit power (dBm)

G_t, G_r = transmitter and receiver antenna gains (dBi)

L_p = free-space path loss (dB)

L_{sys} = system losses (cables, filters, etc.)

Free-Space Path Loss Formula: $L_p(\text{dB}) = 20\log(d) + 20\log(f) + 32.44$

where d = distance in km, f = frequency in MHz.

For PAUSAT at 550 km and $f = 2250$ MHz: $L_p = 20\log(550) + 20\log(2250) + 32.44 \approx 155.2$ dB

This showed me why high-gain ground antennas and sensitive receivers are essential for reliable CubeSat communications.

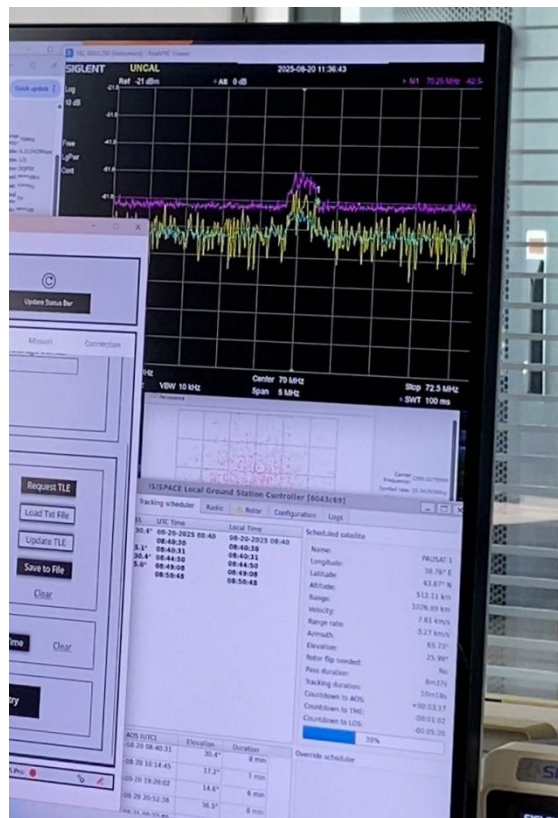


Figure X. SVNA display showing PAUSAT-1 signal peak with Doppler shift visible as frequency drift.

Temporary Clean Room for RAFS

After the ground station tour, we visited the temporary clean room where engineers were integrating RAFS (Rubidium Atomic Frequency Standard) components. In particular, the On-Board Computer (OBC) and ADCS modules were being tested. The OBC handled telemetry routing and fault detection, while the ADCS system, consisting of magnetorquers and reaction wheels, was responsible for attitude stabilization.

Seeing the subsystems up close helped me connect theoretical block diagrams from my lectures to real hardware. For example, an ADCS wheel typically has angular momentum storage of ~ 0.02 – 0.05 Nms, which at max torque (~ 0.01 Nm) provides slew rates suitable for CubeSat-scale pointing requirements.



Figure. PAUSAT Model



Figure. RAFS OBC and ADCS modules being integrated in a temporary clean room.

Introduction to STK with PAUSAT-1

Before working on PAUSAT-1 directly, we first needed to familiarize ourselves with ANSYS Systems Tool Kit (STK). After downloading and installing the software, we dedicated two days of the week to going through the official STK tutorials and video guides provided on ANSYS's website. These materials explained the fundamentals of the software: how to create new scenarios, insert satellites and ground facilities, adjust orbital parameters, and generate visualizations such as ground tracks and access windows.

This initial deep dive was very valuable, as it gave me the confidence to move from simply following instructions to actually setting up my own scenarios. Once we had completed these

tutorials and gained a basic understanding of STK's features, we began applying what we learned to a real case studies..

We began working with ANSYS STK by importing Two-Line Element (TLE) data for PAUSAT-1:

```
62653U 25009AW 25244.91059407 .00008191 00000-0 37959-3 0 9995
62653 97.4226 323.8499 0000594 181.8974 178.2259 15.20337230 34937
```

From the second line, I extracted key orbital parameters:

Inclination (i) = 97.42° → sun-synchronous orbit.

RAAN = 323.85°.

Eccentricity (e) = 0.0000594 (almost circular).

Mean Motion = 15.203 rev/day → orbital period ~95 minutes.

Orbital period formula:

$T = 24 \times 60n$

where $n = 15.203$ rev/day.

$T = 1440 \times 15.203 \approx 94.7$ min

This matched STK's simulation and confirmed PAUSAT's low Earth orbit.

Besides working on PAUSAT-1, we also performed a lifetime analysis of SharjahSat-1 using STK. The goal was to estimate how long the satellite would remain in orbit before atmospheric drag caused re-entry.

In STK, we used the Orbit Lifetime Tool, which calculates orbital decay based on altitude, ballistic coefficient, and atmospheric density models.

Step 1 - Input Parameters:

Table 2.5: SharjahSat-1 configuration and orbital parameters

Parameter	Value
Launch Date	Fourth Quarter of 2022
Mass	4.107 kg
Drag Area	0.012 m ²
Area Exposed to Sun	0.02 m ²
C _D	2.2
C _R	1
Drag Area/Mass	0.0029218
Area Exposed to Sun/Mass	0.0048697

Step 2 - Ballistic Coefficient:

$$B = \frac{m}{C_d \cdot A}$$

where:

m = satellite mass (kg)

C_d = drag coefficient

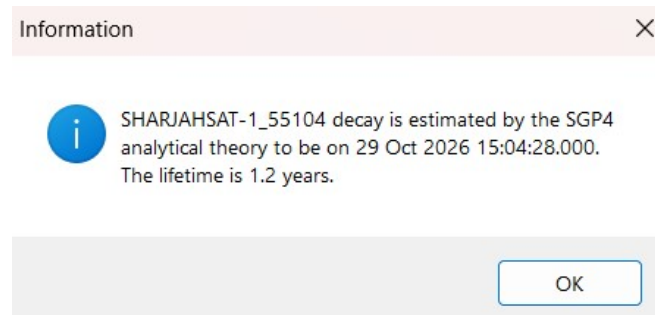
A = reference cross-sectional area (m²)

For SharjahSat:

$$B=4.107/(2.2 \times 0.012) \approx 155.57 \text{ kg/m}^2$$

Step 3 - Lifetime Estimate:

STK uses atmospheric density models (like NRLMSISE-00) to propagate decay. With $B \approx 155.57$, the simulation predicted a lifetime of **~1.2 years** at 550 km, depending on solar activity.



Predicted orbital lifetime of SharjahSat using STK lifetime tool.

FOURTH WEEK OF THE INTERNSHIP

Step 1 – Creating a Customized Scenario in STK

At the start of the week, I recreated the Pro Version workflow from Harding Labs. I began by opening STK Pro and creating a new scenario with a timeline covering 48 hours (29–30 August 2025). This ensured that the software would compute multiple orbital passes for my satellites.

Orbit parameters chosen: Altitude = 550 km (LEO), inclination = 50° , circular orbit.

Step 2 – Inserting Customized Satellites

Instead of using real-world satellites, I inserted three customized satellites (Satellite1, Satellite2, Satellite3). Each was defined with the same orbital altitude and inclination, but their RAANs (Right Ascension of Ascending Node) were shifted by 120° to space them evenly:

Satellite1 $\rightarrow$ RAAN = 0°

Satellite2 $\rightarrow$ RAAN = 120°

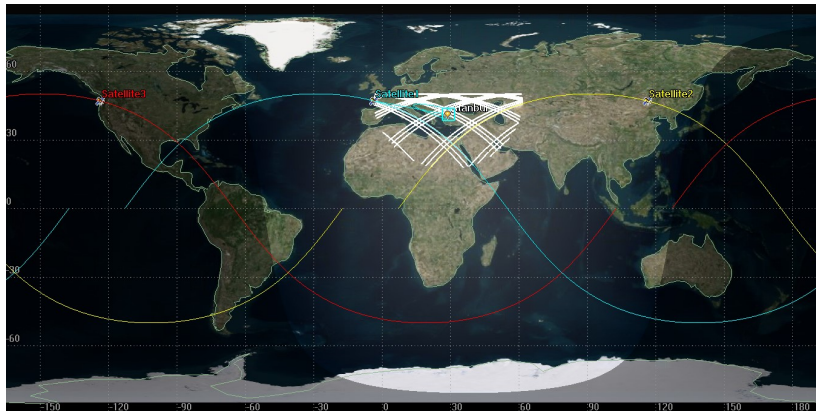
Satellite3 $\rightarrow$ RAAN = 240°

Why RAAN spacing?

RAAN separation ensures that the satellites are distributed evenly across orbital planes. This maximizes revisit frequency and reduces coverage gaps.

Formula for ground track spacing: $\Delta\Omega = 360^\circ/N$

Where $N=3$ (number of satellites), so $\Delta\Omega=120$.



2D world map with three orbital planes over Istanbul

Step 3 – Defining the Target (Place: Istanbul)

Next, I created a Place object at the geographic coordinates of Istanbul. This served as the observation target.

Latitude = 41.0082° N

Longitude = 28.9784° E

Elevation = 10 m

Step 4 – Attaching Sensors to Satellites

For each satellite, I attached a conic sensor with the following parameters:

Type: Simple Conic

Cone Half Angle = 5°

Pointing = Directed at Istanbul

Swath Width Formula:

$$\text{Swath} = 2 \times H \times \tan\left(\frac{\theta}{2}\right)$$

Where:

H=550km (altitude)

$\theta=5^\circ$

$$\text{Swath} = 2 \times 550,000 \times \tan(2.5^\circ) \approx 48,000 \text{ m} = 48 \text{ km}$$

Thus, each satellite had ~48 km ground swath coverage centered on Istanbul.



3D globe screenshot showing the sensor cone over Istanbul.

Step 5 - Computing Access Reports

Using the Access Tool in STK, I generated reports for each satellite to Istanbul. These reports included:

Start time (AOS: Acquisition of Signal)

Stop time (LOS: Loss of Signal)

Duration of access (sec)

Definitions:

AOS (Acquisition of Signal): When satellite rises above the minimum elevation angle and becomes visible to the ground target.

LOS (Loss of Signal): When the satellite sets below the elevation mask and the line-of-sight is lost.

Pass Duration Formula:

$$t_{pass} \approx \frac{2}{\omega} \cos^{-1} \left(\frac{R_E}{R_E + h} \cos E_{min} \right)$$

Where:

ω = Earth's angular velocity ($\sim 15^\circ/\text{min}$ for LEO)

$R_E=6371\text{km}$, $h=550\text{km}$, $E_{min}=10^\circ$

$t_{pass} \approx 8\text{-}12$ minutes

This matched the durations in my access reports ($\sim 700\text{--}800$ sec).

Civil Air Patrol Use Only
Satellite-Satellite3-Sensor-Sensor3-To-Place-Istanbul: Access Summary Report

Sensor3-To-Istanbul					
	Access	Start Time (UTC/G)		Stop Time (UTC/G)	Duration (sec)
	1	29 Aug 2025	15:34:27.617	29 Aug 2025 15:37:35.031	187.415
	2	29 Aug 2025	17:09:32.976	29 Aug 2025 17:22:06.661	753.684
	3	29 Aug 2025	18:49:47.704	29 Aug 2025 19:03:17.610	809.907
	4	29 Aug 2025	20:31:41.513	29 Aug 2025 20:44:32.184	770.670
	5	29 Aug 2025	22:13:33.355	29 Aug 2025 22:26:29.578	776.223
	6	29 Aug 2025	23:54:42.905	30 Aug 2025 00:08:14.874	811.969
	7	30 Aug 2025	01:36:01.607	30 Aug 2025 01:48:04.467	722.860
Global Statistics					
Min Duration	1	29 Aug 2025	15:34:27.617	29 Aug 2025 15:37:35.031	187.415
Max Duration	6	29 Aug 2025	23:54:42.905	30 Aug 2025 00:08:14.874	811.969
Mean Duration					690.390
Total Duration					4832.727

4 Sep 2025 14:04:24

Civil Air Patrol Use Only
Satellite-Satellite1-Sensor-Sensor1-To-Place-Istanbul: Access Summary Report

Sensor1-To-Istanbul				
	Access	Start Time (UTCG)	Stop Time (UTCG)	Duration (sec)
1	29 Aug 2025 09:28:53.478	29 Aug 2025 09:41:37.056	763.578	
2	29 Aug 2025 11:12:39.548	29 Aug 2025 11:17:34.294	294.746	
3	29 Aug 2025 23:36:23.788	29 Aug 2025 23:42:16.624	352.837	
4	30 Aug 2025 01:12:49.112	30 Aug 2025 01:25:39.489	770.377	
5	30 Aug 2025 02:53:19.634	30 Aug 2025 03:06:46.765	807.132	
6	30 Aug 2025 04:35:17.853	30 Aug 2025 04:48:05.482	767.629	
7	30 Aug 2025 06:17:05.185	30 Aug 2025 06:30:05.908	780.723	
8	30 Aug 2025 07:58:11.989	30 Aug 2025 08:11:43.749	811.760	
Global Statistics				
Min Duration	2	29 Aug 2025 11:12:39.548	29 Aug 2025 11:17:34.294	294.746
Max Duration	8	30 Aug 2025 07:58:11.989	30 Aug 2025 08:11:43.749	811.760
Mean Duration				668.598
Total Duration				5348.782

4 Sep 2025 14:04:23

Civil Air Patrol Use Only
Satellite-Satellite2-Sensor-Sensor2-To-Place-Istanbul: Access Summary Report

Sensor2-To-Istanbul				
	Access	Start Time (UTCG)	Stop Time (UTCG)	Duration (sec)
	1	29 Aug 2025 09:06:19.412	29 Aug 2025 09:18:32.535	733.123
	2	29 Aug 2025 10:46:16.941	29 Aug 2025 10:59:48.556	811.615
	3	29 Aug 2025 12:28:05.084	29 Aug 2025 12:40:59.502	774.418
	4	29 Aug 2025 14:10:00.934	29 Aug 2025 14:22:53.149	772.216
	5	29 Aug 2025 15:51:13.864	29 Aug 2025 16:04:44.659	810.795
	6	29 Aug 2025 17:32:26.830	29 Aug 2025 17:44:52.108	745.278
	7	30 Aug 2025 07:38:53.330	30 Aug 2025 07:46:25.997	452.666
Global Statistics				
Min Duration	7	30 Aug 2025 07:38:53.330	30 Aug 2025 07:46:25.997	452.666
Max Duration	2	29 Aug 2025 10:46:16.941	29 Aug 2025 10:59:48.556	811.615
Mean Duration				728.587
Total Duration				5100.110

STK Access Report for SatelliteX-to-Istanbul showing pass durations and statistics.

Step 6 – Results and Revisit Frequency

From the reports:

Satellite1 → 8 passes in 48 hrs, mean duration ≈ 668 sec.

Satellite2 → 7 passes in 48 hrs, mean duration ≈ 728 sec.

Satellite3 → 7 passes in 48 hrs, mean duration ≈ 690 sec.

Total passes = 22 in 48 hrs → 11 passes/day.

For comparison:

A single satellite gave ~ 5 – 6 passes/day.

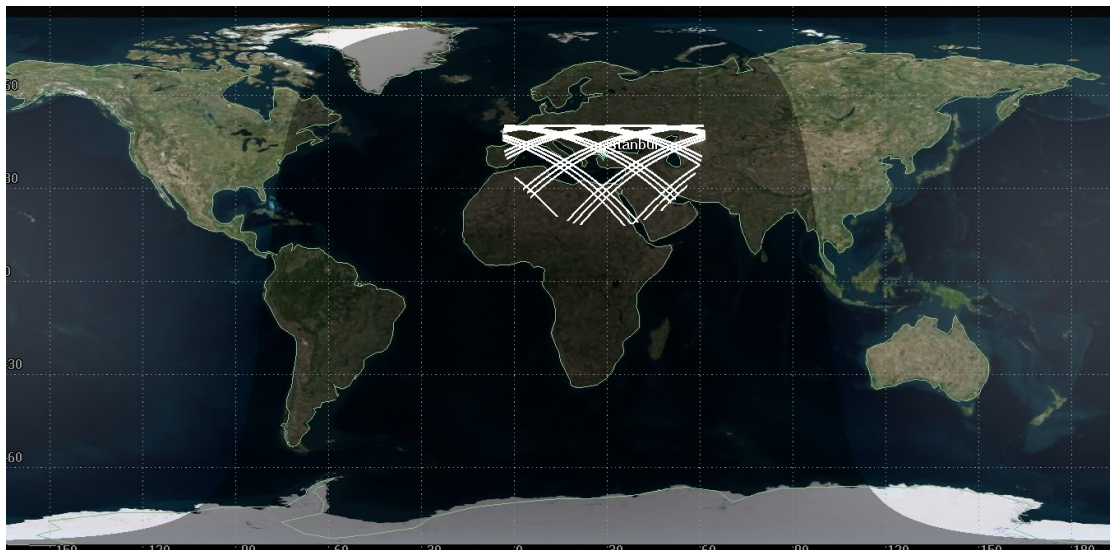
A 3-satellite constellation gave ~ 11 passes/day.

Revisit Improvement Formula:

$$\text{Revisit Rate}_{\text{constellation}} = N \times \text{Revisit Rate}_{\text{single}}$$

Here: $N=3$, Single = ~ 6 /day → Predicted ≈ 18 /day.

In practice, I observed ~ 11 /day because access opportunities overlap. This discrepancy taught me that theoretical multiplication works only for widely separated orbital planes, and STK is essential to simulate real-world geometry.



2D access map showing dense white coverage lines over Istanbul.

Reflection

This week taught me how to use STK Pro's professional analysis workflow for constellation design and revisit analysis. I learned how RAAN separation, swath geometry, and access reporting come together to answer mission-critical questions like "How often can we observe a region?" Applying this process to Istanbul gave me hands-on experience in designing and evaluating customized constellations, which I believe will be highly valuable for Earth observation and disaster monitoring projects in my future career.

References:

- **ITU USTTL (İstanbul Teknik Üniversitesi, Uzay Sistemleri Tasarım ve Test Laboratuvarı).**
Laboratory facilities, CubeSat missions (İTÜpSAT-1, TURKSAT-3USAT, BeEagleSat, Havelsat, UBAKUSAT, ASELSAT, PAUSAT-1, RAFS), and ground station details. Internal internship documents and USTTL official publications.
- **UHUZAM (İTÜ CSCRS – Satellite Communications and Remote Sensing Center).**
Ground station specifications: 13 m antenna, VSAT terminals, ~3000 km X-band coverage footprint. Technical documentation provided by İTÜ.
- **OroraTech (2023–2025).**
SAFIRE Gen-2 Sensor Specifications. Company presentations, technical datasheets, and ESA InCubed project materials. Munich, Germany.
- **ESA InCubed Program.**
Forest-3 satellite project information including payload integration of OroraTech SAFIRE Gen-2 thermal sensor. European Space Agency, 2022–2025.
- **Leonardo Electronics.**
Condor HD Dual-Band Infrared Camera datasheet and technical specifications. Leonardo S.p.A., 2023.
- **SDC Inc.**
BIRD 640 VOx Uncooled Microbolometer Sensor datasheet and technical specifications. SDC Technologies, 2023.
- **Iridian Spectral Technologies.**
Dual-band and tri-band mosaic filters product datasheets. Iridian Optics, 2022–2024.
- **Edmund Optics.**
ZnSe/ZnS plano-convex lens catalog for MWIR and LWIR applications. Edmund Optics, 2023.
- **Simera Sense.**
TriScape100 multispectral optical payload datasheet. Simera Sense, 2023.
- **Harding, J. (2023).**
Calculating Satellite Visitation Frequency with STK (Pro Version). Harding Labs Blog. Retrieved from: <https://blog.hardinglabs.com/private-observer.html>

- **AGI/ANSYS (2025).**

Systems Tool Kit (STK) User Documentation and Tutorials. Accessed from ANSYS official learning resources.

- **N2YO Satellite Tracking Website.**

Used for cross-checking orbital elements and PAUSAT-1 visibility windows. Retrieved from: <https://www.n2yo.com>

- **NRLMSISE-00 Atmospheric Model.**

Used within STK for orbital lifetime estimation of SharjahSat and PAUSAT-1. Naval Research Laboratory, 2000.

- **Friis, H. T. (1946).**

“A Note on a Simple Transmission Formula.” *Proceedings of the IRE*, 34(5): 254–256. [For link budget calculations in Week 3 & 4].

- **ITU-R P.618-13 (2017).**

Propagation data and prediction methods required for the design of Earth-space telecommunication systems. International Telecommunication Union.